Spatial ML - class 3 - obtaining information from the environment

Total points 100/100

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1. What does a spatial weights matrix W primarily define? * 10/10

- (●) The pattern of spatial relationships between units
- ( ) The magnitude of spatial autocorrelation
- ( ) The distribution of the dependent variable
- ( ) The strength of regression coefficients

2. In a queen contiguity matrix, two regions are neighbours when they: * 10/10

- ( ) Are within a fixed distance threshold
- ( ) Share a border
- (●) Share a border or a corner
- ( ) Belong to the same administrative level

3. After standardization of matrix (style = "W"), which property holds? 10/10

- ○ The matrix becomes symmetric
- ○ Each column sums to one
- ○ All weights become identical
- ◉ Each row sums to one

4. If a region has 4 neighbours in a row-standardised matrix, each neighbour receives weight: *10/10

- ◉ 0.25
- ○ 1
- ○ Depends on the neighbour's value
- ○ 4

5. The spatial lag represents: * 10/10

- ( ) The variance of neighbouring observations
- ( ) The difference from the global mean
- (•) The weighted average of neighbouring values
- ( ) The sum of all neighbouring values

6. Which statement about KNN matrices is correct? * 10/10

- ( ) They are always symmetric
- (•) Each unit initially has exactly k neighbours
- ( ) They guarantee a connected network
- ( ) They automatically sum to one by row

7. Why can different spatial weights matrices lead to different model results? * 10/10

- ○ Because spatial data are random
- ○ Because regression ignores spatial structure
- ◉ Because W encodes assumptions about interaction
- ○ Because the spatial lag is constant

8. Fragmentation in a spatial weights matrix refers to: * 10/10

- ○ Very large weight values
- ○ Units connected to all others
- ◉ Units with no neighbours
- ○ Presence of negative autocorrelation

9. Inverse distance matrices can be computationally demanding because they are often: *10/10

- ( ) Symmetric by construction
- ( ) Based only on polygon data
- ( ) Sparse and highly diagonal
- (•) Dense with many non-zero entries

10. Radial zoning is particularly useful for point data because: 10/10

- ( ) Points naturally share borders
- (•) Interactions are typically local and continuous
- ( ) It guarantees matrix symmetry
- ( ) It eliminates spatial dependence